

Mikołaj Karkowski

Full-stack .NET Engineer

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PROFILE

Full-stack .NET engineer with 7+ years of commercial experience building web and desktop products for complex, data-intensive domains. I turn specialized requirements into maintainable software—from real-time Blazor interfaces and ASP.NET Core services to WPF clients and applied AI systems. I value clear boundaries, strong typing, practical automation and close collaboration with users.

CORE SKILLS

Languages & runtime

C#, .NET 10, SQL

Web

ASP.NET Core, Blazor
WebAssembly, Angular, REST APIs,
SignalR

Desktop

WPF, MVVM, .NET MAUI Hybrid

Data

EF Core, PostgreSQL, SQL Server,
SQLite

AI

Microsoft.Extensions.AI, OpenRouter,
GitHub Models, Ollama, YOLO

Engineering

DDD, CQRS, modular monoliths,
Docker, .NET Aspire, OpenTelemetry,
GitHub Actions

Testing

xUnit, Moq, WebApplicationFactory

EDUCATION

Master of Computer Science

2021–2022
Silesian University of Technology

Bachelor of Automation and Robotics

2016–2021
Silesian University of Technology

LANGUAGES

Polish — native
English — C1

INTERESTS

Photography, hiking, travel

EXPERIENCE

Software Developer

Apr 2019 — present

Graw / Goldschmidt Group

Design, develop and maintain web and desktop .NET products for railway measurement, inspection and navigation systems. Collaborate directly with clients to shape requirements and deliver effective solutions.

- Built Blazor WebAssembly and .NET MAUI Hybrid interfaces presenting live camera feeds, laser measurements and equipment status with role-based UI access.
- Created WPF tools for visualizing measurements, reviewing rail defects, displaying equipment diagnostics and working with video-backed image data.
- Developed an AI-assisted train-navigation algorithm combining video, computer vision, rail infrastructure and operational data as an alternative where GPS is unreliable.
- Built a rail-object detection tool and trained YOLO models for finding damage on tracks and sleepers.

Engineering Intern

Apr 2018 — Jun 2018

Bobson

Programmed Arduino boards, tested CAN networks and operated milling equipment.

Junior Software Developer Intern

Apr 2015

QNT Systems

Developed a responsive web application using Bootstrap.

SELECTED PROJECTS

FinanceManager

GitHub

Open-source personal finance platform

Full-stack .NET 10 product for managing cash, stocks and bonds with analytics, live market data, CSV workflows and AI-generated insights. Built as a layered modular monolith with a pure domain layer, typed HTTP clients, background channels, SignalR, JWT authentication and OpenTelemetry.

Kotlet

GitHub

Connected home-kitchen workspace

Recipe, pantry, meal-planning and shopping platform with a shared ingredient catalogue. Includes 28-day planning, nutrition and cost calculations, public recipe sharing, image workflows and AI-assisted imports from video sources.